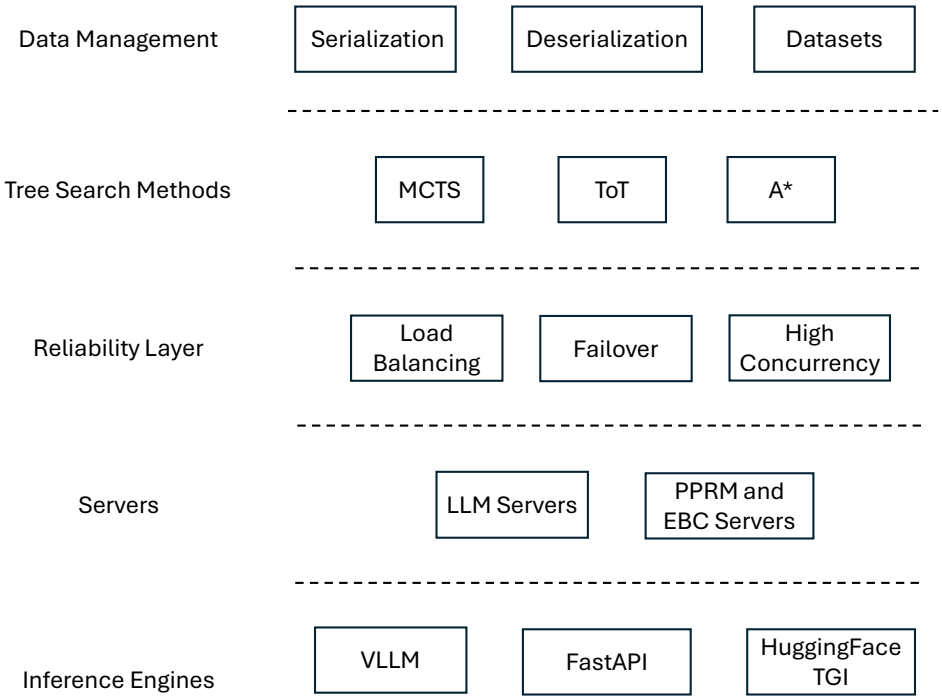

Figure A3: Architecture design of `Berry-Tree`

D.4 Preliminary Performance Evaluation

In a preliminary performance evaluation, we utilize 16 A100 GPUs to run the LLaMA3.1-8B-instruct model for large-scale inference tasks, while 4 A100 GPUs are used to run the Gemma2-2B-it model as PPRM servers. The test dataset consists of 1319 GSM8K test samples.

We conduct 16 rollouts to parallelize the inference tasks of `LLaMA-Berry` via `Berry-Tree`. The results indicate that the total inference time is 1 hour and 25 minutes, with an average inference time of approximately 3.87 seconds per sample. These results demonstrate a strong parallel inference capabilities of `Berry-Tree` under the given hardware configuration.

E Scaling Study on Inference-time Token Overheads

Comparing LLaMA-3-8B-Instruct and LLaMA-3.1-8B-Instruct across GSM8K, GSMHARD, and MATH500, LLaMA-3.1-8B-Instruct consistently consumes more tokens across all categories—Solutions, Critiques, and Overall. This Figure A4 result reflects LLaMA-3.1-8B-Instruct's tendency to generate more detailed and comprehensive outputs with increased overhead. While this likely improves solution quality, it also introduces greater resource demands and variability in token usage, highlighting a trade-off between accuracy and computational resources. LLaMA-3.1-8B-Instruct is thus better suited for tasks prioritizing precision over speed. As shown in Figure A5, Token overheads during inference also scale with task difficulty across different olympiad-level benchmarks with LLaMA-3.1-8B-Instruct. AIME2024 exhibits the highest token consumption with significant variability, reflecting the complexity of its solution paths. In contrast, GPQA Diamond shows lower overall overhead, while AMC2023 falls in between, with moderate token consumption and less variability than AIME2024 but still notable.

F Future Work

`LLaMA-Berry` holds significant potential for further development. First, we plan to enhance its multimodal capabilities, enabling it to better handle complex problems like VQA and mathematic geometric problems that require visual, auditory, or even tactile perception. For instance, in tasks involving visual reasoning, such as geometric problems, `LLaMA-Berry` could be extended to integrate image recognition and analysis capabilities, thereby assisting in solving challenges related to spatial relationships and shape recognition. Furthermore, we aim

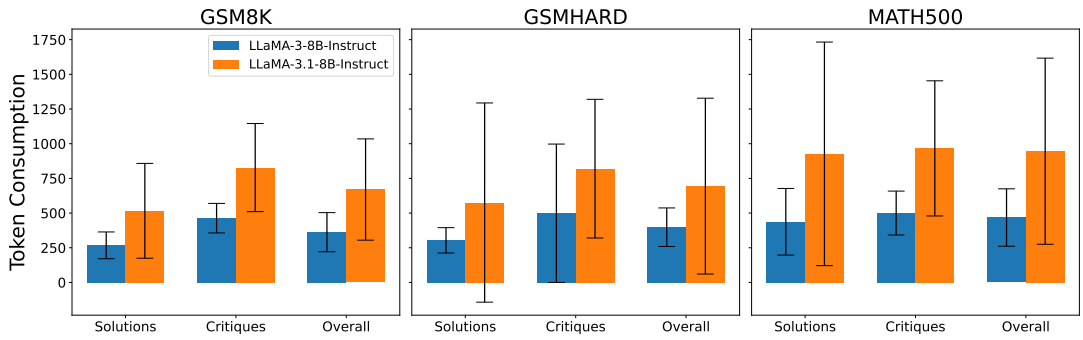

Figure A4: Average token consumption comparison across datasets, error bar stands for standard deviation.

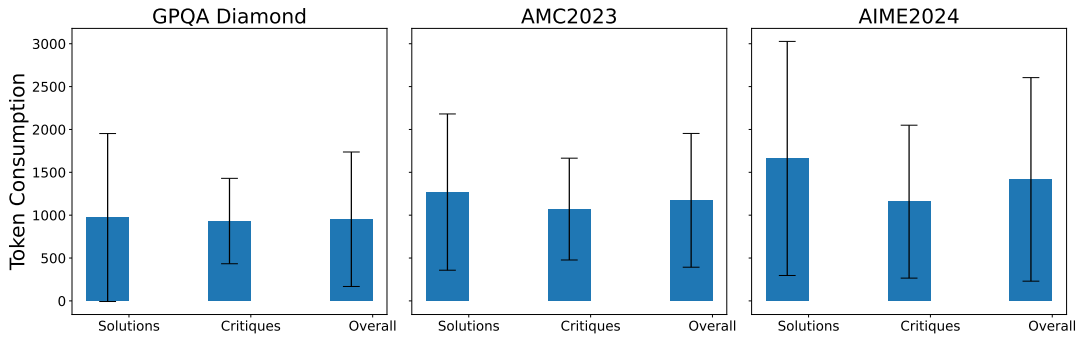

Figure A5: Average token consumption for LLaMA-3.1-8B-Instruct across olympiad-level datasets, error bar stands for standard deviation.

to generalize its application across other scientific domains, improving its performance in disciplines such as physics, chemistry, and biology. In physics, for example, it could be utilized to address complex micro- and macroscopic dynamics problems; in chemistry, it may assist in molecular structure prediction and drug design; and in biology, it could potentially be used for genome analysis and disease prediction. Moreover, `LLaMA-Berry` can also offer technical support for other interdisciplinary fields, such as meteorology, environmental science, and materials science, by integrating multimodal data to enhance predictive accuracy. At the same time, we will focus on how `LLaMA-Berry` can further enhance AI safety. For instance, `LLaMA-Berry` could be leveraged to design more robust safety and risk assessment mechanisms. Besides, the generation of responses could be guided by a safety-trained PPRM to produce outputs with higher safety standards. These advancements not only pave the way for broader scientific applications of `LLaMA-Berry` but also offer new possibilities for enhancing AI safety and promoting its widespread adoption.

G Case Study

The prompts utilized in `LLaMA-Berry` for the LLaMA-3.1-8B-Instruct model are presented in Figure A6. Additionally, Figure A7 provides a detailed breakdown of problem-solving examples derived from the GSM8K dataset.

H Convergence Analysis of the Enhanced Borda Count (EBC) Method

In this appendix, we present a formal discussion on how the quantile scores, as evaluated by the Enhanced Borda Count (EBC) method, converge to the true quantile scores of solutions within the actual quality distribution as the number of samples increases.

H.1 Definitions and Assumptions

Solution Set. Let $A = \{a_1, a_2, \ldots, a_n\}$ be a finite set of n solutions (answers).

True Quantile Scores. Each solution a_i has a true quality score $Q^*(a_i) \in \mathbb{R}$, drawn from a continuous distribution P_Q.

True Ranking. The true ranking R^* is induced by ordering the solutions in decreasing order of their

Component	Details
Generate the first solutions	Please write an answer for this question. The Answer should format as '#### Reasoning Process ... #### Verification ... #### Final Answer ...'. The Final Answer should format as \boxed{Final Answer} Let's think step by step
Generate Critique	Analyze this weak Answer, write a strict Critic/Reflection for error re-correct and Hints/Guidelines for maximum improvement. Let's think step by step.
Rewriting	Please write a better answer for this question refer to the comments. The Answer should format as '#### Reasoning Process ... #### Verification ... #### Final Answer ...'. The Final Answer should format as \boxed{Final Answer} Let's think step by step
Instruction PPRM	Problem: ... First Answer: ... Second Answer: ... Is First Answer better than Second Answer?

Figure A6: Prompts for LLaMA-3.1-8B-Instruct

Component	Details
Problem Statement	Lee used to be able to run the 400-meter hurdles two seconds faster than Gerald would run the 400-meter hurdles. But Gerald changed his diet, which improved his speed by 10%. If Lee runs the 400-meter hurdles in 38 seconds, how fast can Gerald, with his improved diet, run the 400-meter hurdles, in seconds?
First Attempt	**Reasoning Process:** 1. Gerald's original time: 40 seconds (Lee's time + 2 seconds). 2. Calculate improvement: 10% of 40 seconds = 4 seconds. 3. New time: 40 + 4 = 44 seconds. ERROR
	Verification: Calculated time based on Lee's current time and original difference. Final Answer: 44 seconds.
Critique of First Attempt	**Reflection:** 1. Lack of clear explanation. 2. Inaccurate percentage increase representation. 3. Incorrect verification. 4. Insufficient mathematical notation
	Error Re-correction: 1. Gerald's original time: 40 seconds. 2. New time: $40 - 4 = 36$ seconds
Improved Answer	**Reasoning Process:** 1. Lee runs in 38 seconds. 2. Gerald's original time: 40 seconds. 3. Improvement: 4 seconds. 4. New time: 36 seconds. CORRECT
	Final Answer: 36 seconds.

Figure A7: Problem-solving example

true quantile scores $Q^*(a_i)$.

True Pairwise Preference Probabilities. The true probability that solution a_i is preferred over a_j is defined as:

$$P^*(a_i \succ a_j) = \mathbb{P}(Q^*(a_i) > Q^*(a_j)) + \frac{1}{2}\mathbb{P}(Q^*(a_i) = Q^*(a_j)). \tag{A1}$$

Given the continuity of P_Q, we have $\mathbb{P}(Q^*(a_i) = Q^*(a_j)) = 0$, so:

$$P^*(a_i \succ a_j) = \begin{cases} 1, & \text{if } Q^*(a_i) > Q^*(a_j), \\ 0, & \text{if } Q^*(a_i) < Q^*(a_j). \end{cases} \tag{A2}$$

Estimated Preference Probabilities. For T independent samples, the estimated preference proba-